Math for CS & AI: Homework 3

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Problem 15

A polynomial rolling hash function is defined as

$$h_{\alpha,p}\left((x_i)_{i=0}^{\ell-1}\right) = \left(\sum_{i=0}^{\ell-1} \alpha^{\ell-1-i} x_i\right) \bmod p,$$

where p is a prime number with $p > u$.

Based on $h_{\alpha,p}$, we define two hash functions that map from $[u]^\ell$ to a smaller hash space $\mathbb{Z}_m$, where m can be much smaller than p and may not be prime.

$$h^{(1)}_{\alpha,p,m}\left((x_i)_{i=0}^{\ell-1}\right) := h_{\alpha,p}\left((x_i)_{i=0}^{\ell-1}\right) \bmod m,$$

$$h^{(2)}_{\alpha,a,b,p,m}\left((x_i)_{i=0}^{\ell-1}\right) := \left(\left(a \cdot h_{\alpha,p}\left((x_i)_{i=0}^{\ell-1}\right) + b\right) \bmod p\right) \bmod m.$$

Let $\mathcal{H}^{(1)}_{p,m}$ and $\mathcal{H}^{(2)}_{p,m}$ be the corresponding hash families.

$$\mathcal{H}^{(1)}_{p,m} := \left\{h^{(1)}_{\alpha,p,m} : \alpha \in \mathbb{Z}_p \setminus \{0\}\right\},$$

$$\mathcal{H}^{(2)}_{p,m} := \left\{h^{(2)}_{\alpha,a,b,p,m} : \alpha \in \mathbb{Z}_p \setminus \{0\}, a \in \mathbb{Z}_p \setminus \{0\}, b \in \mathbb{Z}_p\right\}.$$

For each of the two hash families, try to find a universal constant $c > 0$ such that the following holds. For any integers p, m, L satisfying $u < m \le p$ and $L \ge 1$, and for any two distinct strings $(x_i)_{i=0}^{\ell-1}$ and $(x'_i)_{i=0}^{\ell'-1}$ with $\ell, \ell' \le L$,

$$\Pr\left[h\left((x_i)_{i=0}^{\ell-1}\right) = h\left((x'_i)_{i=0}^{\ell'-1}\right)\right] \le \frac{cL}{m},$$

where h is sampled uniformly at random from the corresponding hash family. If you cannot find such a constant, give a counterexample.

Answer:

Define events $A^{(1)}$ and $A^{(2)}$ as the collision event that $h\left((x_i)_{i=0}^{\ell-1}\right) = h\left((x'_i)_{i=0}^{\ell'-1}\right)$, where the hash function h is sampled uniformly at random from the corresponding hash family.

For distinct sequences $(x_i)_{i=0}^{\ell-1}$ and $(x'_i)_{i=0}^{\ell'-1}$, we define $\Delta(\alpha) = \left[h_{\alpha,p}\left((x_i)_{i=0}^{\ell-1}\right) - h_{\alpha,p}\left((x'_i)_{i=0}^{\ell'-1}\right)\right] \bmod p$.

For the hash family $\mathcal{H}^{(1)}_{p,m}$, consider the following construction:

- Let $p = k \cdot m + r$, $u = r + 1$, and $L = 2$, where $r = m - 2$ and k is an arbitrary positive integer.
- Consider the length-2 strings $(1, 1)$ and $(1, r + 1)$. Their corresponding polynomial hash values are $(\alpha + 1) \bmod p$ and $(\alpha + r + 1) \bmod p$, respectively.
- For all α satisfying $p - r - 1 \leq \alpha \leq p - 2$, the collision condition $\alpha + 1 \equiv \alpha + r + 1 - p \pmod{m}$ holds, meaning the two strings have the same hash value under $h^{(1)}_{\alpha,p,m}$.

Under this construction, the collision probability $\Pr\left[A^{(1)}\right] \geq \frac{m-2}{p-1}$, which implies $c \geq \frac{m(m-2)}{(p-1)L}$. The lower bound will increase while the given p, m changing, therefore, no such universal constant exists for the hash family $\mathcal{H}^{(1)}_{p,m}$.

For the hash family $\mathcal{H}^{(2)}_{p,m}$, a collision can be split into two mutually exclusive cases:

- $\Delta(\alpha) = 0$. Here, $\Delta(\alpha)$ is a polynomial with integer coefficients, and it is a non-zero polynomial since the sequences (x_i) and (x'_i) are distinct. Therefore, the congruence $\Delta(\alpha) \equiv 0 \pmod{p}$ has at most $L - 1$ solutions for $\alpha \in \mathbb{Z}_p \setminus \{0\}$. Thus, the probability of collision in this case is at most $\frac{L-1}{p-1}$.
- $\Delta(\alpha) \neq 0$. This means the hash values $h\left((x_i)_{i=0}^{\ell-1}\right)$ and $h\left((x'_i)_{i=0}^{\ell'-1}\right)$ are distinct integers in $\mathbb{Z}_p$. By Proposition 2.54, which we proved in Problem 14, the probability of collision in this case is at most $\frac{1}{m}$.

If $m = p$, no collision can occur in the second case, which gives $\Pr\left[A^{(2)}\right] \leq \frac{L-1}{p-1} \leq \frac{L}{m}$. Otherwise, $m < p$, and we have $\Pr\left[A^{(2)}\right] \leq \frac{L-1}{p-1} + \frac{1}{m} \leq \frac{L}{m}$. In conclusion, $\Pr\left[A^{(2)}\right] \leq \frac{L}{m}$, which implies that there exists a universal constant $c = 1$ for the hash family $\mathcal{H}^{(2)}_{p,m}$.

Problem 16

Let $R(k, \ell)$ be the least number N such that every red/blue coloring of the edges of K_N contains a red K_k or a blue K_ℓ.

(a) Prove that $R(k, \ell)$ exists for all $k, \ell \geq 1$ and satisfies the following recurrence relation:

$$R(k, \ell) \leq R(k-1, \ell) + R(k, \ell-1).$$

(b) Prove the following upper bound on $R(k, k)$ for all $k \geq 2$:

$$R(k, k) \leq \binom{2k-2}{k-1} < 4^k.$$

Answer:

For part (a), in the boundary cases where either $k = 1$ or $\ell = 1$, $R(1, \ell) = R(k, 1) = 1$ because a single vertex K_1 always contains a red K_1 or a blue K_1.

Take any complete graph K_N with $N = R(k-1, \ell) + R(k, \ell-1)$ vertices, and pick an arbitrary vertex $v \in K_N$. Split the remaining $N - 1 = R(k-1, \ell) + R(k, \ell-1) - 1$ vertices into two disjoint sets:

- S_R: the set of vertices connected to v by a red edge;
- S_B: the set of vertices connected to v by a blue edge.

By the pigeonhole principle, at least one of the following must hold: either $|S_R| \geq R(k-1, \ell)$, or $|S_B| \geq R(k, \ell-1)$. We analyze the two cases separately:

- If $|S_R| \geq R(k-1, \ell)$, by the definition of $R(k-1, \ell)$, the induced subgraph on S_R must contain either a red K_{k-1} or a blue K_ℓ. If there is a blue K_ℓ, the condition is already satisfied. If there is a red K_{k-1}, adding v forms a red K_k, satisfying the condition.
- Otherwise, $|S_B| \geq R(k, \ell-1)$. By the definition of $R(k, \ell-1)$, the induced subgraph on S_B must contain either a red K_k or a blue $K_{\ell-1}$. If there is a red K_k, the condition is already satisfied. If there is a blue $K_{\ell-1}$, adding v forms a blue K_ℓ, satisfying the condition.

Therefore, any red/blue coloring of K_N contains a red K_k or a blue K_ℓ, so $R(k, \ell)$ exists and $R(k, \ell) \leq R(k-1, \ell) + R(k, \ell-1)$.

For part (b), we use mathematical induction to prove the upper bound $R(k, \ell) \leq \binom{k+\ell-2}{k-1}$ for all $k, \ell \in \mathbb{N}^+$. We perform induction on $s = k + \ell$:

- When $s = 2$, we have $k = \ell = 1$. Here $R(1, 1) = 1 \leq \binom{1+1-2}{1-1}$, so the inequality holds. For the boundary cases where either $k = 1$ or $\ell = 1$, $R(1, \ell) = R(k, 1) = 1$, so the inequality holds for all boundary cases.
- Assume that for all integers $k', \ell' \geq 1$ with $k' + \ell' < s$, the inequality $R(k', \ell') \leq \binom{k'+\ell'-2}{k'-1}$ holds. Then consider any $k, \ell \geq 1$ with $k + \ell = s \geq 3$. By the relation from part (a): $R(k, \ell) \leq R(k-1, \ell) + R(k, \ell-1) \leq \binom{k+\ell-3}{k-2} + \binom{k+\ell-3}{k-1} = \binom{k+\ell-2}{k-1}$.

Therefore, this upper bound holds for all $k, \ell \in \mathbb{N}^+$, which implies $R(k, k) \leq \binom{2k-2}{k-1}$ for the case that $k = \ell$.

Since $\binom{2k-2}{k-1} < \sum_i \binom{2k-2}{i} = 2^{2k-2} < 4^k$, we can conclude that $R(k, k) \leq \binom{2k-2}{k-1} < 4^k$.

Problem 17

Denote by $R_r^{(m)}(k)$ the least number N such that Theorem 2.62 holds. Prove that

$$R_r^{(m)}(k) > r^{\frac{1}{k}\left(\binom{k}{m}-1\right)}$$

for all $r, k, m \geq 2$.

Answer:

We use the probabilistic method to prove this lower bound, following the same idea as in Theorem 2.64.

Let $N = \left\lfloor r^{\frac{1}{k}\left(\binom{k}{m}-1\right)} \right\rfloor$, and label the elements by $[N] := 1, 2, ..., N$. Color each m-element subset of $[N]$ independently and uniformly at random with one of the r colors.

We first clarify the definition of a monochromatic k-element subset: a k-element subset $S \subseteq [N]$ is monochromatic if every m-element subset of S is assigned the same color.

Let A be the event that there exists a monochromatic k-element subset. For each k-subset $S \subseteq [N]$, define A_S as the event that S is monochromatic. Then

$$A = \bigcup_{|S|=k} A_S.$$

For fixed $S \subseteq [N]$, there are $\binom{k}{m}$ m-element subsets inside S. For each color, the probability that all these m-element subsets are assigned this color is $r^{-\binom{k}{m}}$. Since there are r possible colors and these cases are mutually exclusive, we have:

$$\Pr[A_S] = r^{1-\binom{k}{m}}.$$

By the union bound,

$$\Pr[A] = \Pr\left[\bigcup_{|S|=k} A_S\right] \leq \sum_{|S|=k} \Pr[A_S] = \binom{N}{k} r^{1-\binom{k}{m}}.$$

Using $\binom{N}{k} \leq \frac{N^k}{k!}$ and $N \leq r^{\frac{1}{k}\left(\binom{k}{m}-1\right)}$,

$$\Pr[A] \leq \frac{\left(r^{\frac{1}{k}\left(\binom{k}{m}-1\right)}\right)^k}{k!} \cdot r^{1-\binom{k}{m}} = \frac{1}{k!} < 1.$$

where the last inequality can be proved for $k \geq 2$. This implies that the complement event $\bar{A}$ has positive probability,

$$\Pr[\overline{A}] > 0.$$

Therefore, there exists an r-coloring of all m-element subsets of an N-element set with no monochromatic k-element subset. By the definition of $R_r^{(m)}(k)$ as the least such N where no such coloring exists, we have $R_r^{(m)}(k) > N = \left\lfloor r^{\frac{1}{k}\left(\binom{k}{m}-1\right)} \right\rfloor$. Since $R_r^{(m)}(k)$ is an integer, this implies $R_r^{(m)}(k) > r^{\frac{1}{k}\left(\binom{k}{m}-1\right)}$.

Problem 18

Let $x_1, \ldots, x_n$ be n vectors in $\mathbb{R}^d$ with $\|x_i\|_2 \leq 1$ for all $i \in [n]$. Prove that there exist $\epsilon_1, \epsilon_2, \ldots, \epsilon_n \in \{-1, +1\}$ such that the following holds:

$$\left\| \sum_{i=1}^{n} \epsilon_i x_i \right\|_2 \leq \sqrt{n}.$$

Answer:

Let $x_{i,p}$ denote the value of the p-th coordinate of the vector x_i. By the definition, the condition $\|x_i\|_2 \leq 1$ is equivalent to $\sum_{p=1}^{d} x_{i,p}^2 \leq 1$ for all $i \in [n]$.

Taking the square of both sides of the target inequality, we only need to prove that there exist $\epsilon_1, \epsilon_2, \ldots, \epsilon_n \in \{-1, +1\}$ such that:

$$\sum_{p=1}^{d} \left(\sum_{i=1}^{n} \epsilon_i x_{i,p} \right)^2 \leq n.$$

We sample each $\epsilon_i \in \{-1, +1\}$ independently and uniformly at random. Consider the expectation of the left-hand side of the above inequality:

$$\begin{aligned}
\mathbb{E}\left[\sum_{p=1}^{d} \left(\sum_{i=1}^{n} \epsilon_i x_{i,p} \right)^2 \right] &= \sum_{p=1}^{d} \sum_{i=1}^{n} \sum_{j=1}^{n} x_{i,p} x_{j,p} \mathbb{E}[\epsilon_i \epsilon_j] && \text{(by linearity of expectation)} \\
&= \sum_{i=1}^{n} \sum_{p=1}^{d} x_{i,p}^2 && \big(\text{since } \mathbb{E}[\epsilon_i \epsilon_j] = 0 \text{ for } i \neq j \text{ while } \epsilon_i^2 = 1\big) \\
&\leq n. && \text{(since } \|x_i\|_2 \leq 1\text{)}
\end{aligned}$$

The left-hand side is a non-negative random variable, and its expectation is at most n. This implies that there must exist at least one realization of $\epsilon_1, \epsilon_2, \ldots, \epsilon_n \in \{-1, +1\}$ for which the value of the random variable is no greater than its expectation, which satisfies the required inequality.

Problem 19

Let A and B be two events in a probability space $P = (\Omega, p)$, each with non-zero probability. We say that A attracts B if $\Pr[B \mid A] > \Pr[B]$. That is, A makes B more likely to happen. Conversely, we say that A repels B if $\Pr[B \mid A] < \Pr[B]$. That is, A makes B less likely to happen.

(a) Prove that A attracts B if and only if B attracts A.

(b) In a poker game of two players, each player samples a random hand of 5 cards from a standard deck of 52 cards. Let A be the event that the first player has a royal flush, and B be the event that the second player has a royal flush. Here, a royal flush means you get a hand of A, K, Q, J, 10 of the same suit (e.g., Spade). There are only 4 possible royal flushes in a deck. Prove that A attracts B.

Answer:

For part (a), by the definition of conditional probability, we have the following equation:

$$\Pr[A \cap B] = \Pr[A] \cdot \Pr[B \mid A] = \Pr[B] \cdot \Pr[A \mid B].$$

Since $\Pr[A] > 0$ and $\Pr[B] > 0$ by the problem's precondition, we can rearrange the equation to get:

$$\frac{\Pr[B \mid A]}{\Pr[B]} = \frac{\Pr[A \mid B]}{\Pr[A]}.$$

If A attracts B, which means $\frac{\Pr[B|A]}{\Pr[B]} > 1$. From the equality above, we immediately get $\frac{\Pr[A|B]}{\Pr[A]} > 1$, which means B attracts A by definition.

The converse direction follows by identical reasoning. Therefore, A attracts B if and only if B attracts A.

For part (b), we first compute the probability of a single player getting a royal flush. There are a total of $\binom{52}{5}$ distinct 5-card hands, and 4 of them are royal flushes. Thus:

$$\Pr[A] = \Pr[B] = \frac{4}{\binom{52}{5}}.$$

Next, we compute the conditional probability $\Pr[B \mid A]$. Given that the first player has a royal flush (event A occurs), there are 47 cards remaining in the deck, which contain exactly 3 complete royal flushes. The number of possible 5-card hands for the second player is $\binom{47}{5}$, so:

$$\Pr[B \mid A] = \frac{3}{\binom{47}{5}}.$$

Expanding the binomial coefficients and we have $3 \cdot \binom{52}{5} > 4 \cdot \binom{47}{5}$. Therefore, $\Pr[B \mid A] > \Pr[B]$, which means A attracts B by definition.

Problem 20

Let A, B, C, D be four events in a probability space $P = (\Omega, p)$. Suppose we know the following:

$$\Pr[A \mid B \cap D] > \Pr[A \mid C \cap D],$$

$$\Pr[A \mid B \cap \bar{D}] > \Pr[A \mid C \cap \bar{D}].$$

Does this imply that $\Pr[A \mid B] > \Pr[A \mid C]$? If yes, prove this inequality. If no, give a counterexample.

Answer:

No, the given conditions do not imply $\Pr[A \mid B] > \Pr[A \mid C]$. We construct a valid counterexample as follows:

- We define a finite probability space with 13 equally likely sample points, corresponding to 13 points distributed across the four quadrants of a coordinate system, where some points are marked.
- The number of points in the 1st, 2nd, 3rd, 4th quadrants are $1, 7, 3, 2$ respectively, with $0, 1, 2, 1$ marked points in each corresponding quadrant.
- The probability space P is defined by selecting a point uniformly at random among these 13 points.
- We define the events:
  - A: the selected point is marked;
  - B: the selected point lies in the 2nd or 3rd quadrant;
  - C: the selected point lies in the 1st or 4th quadrant;
  - D: the selected point lies in the 1st or 2nd quadrant.
- We compute the conditional probabilities by direct calculation:
  - $\Pr[A \mid B \cap D] = \Pr[A \mid \text{2nd quadrant}] = \frac{1}{7} > 0 = \Pr[A \mid \text{1st quadrant}] = \Pr[A \mid C \cap D]$.
  - $\Pr[A \mid B \cap \bar{D}] = \Pr[A \mid \text{3rd quadrant}] = \frac{2}{3} > \frac{1}{2} = \Pr[A \mid \text{4th quadrant}] = \Pr[A \mid C \cap \bar{D}]$.
  - $\Pr[A \mid B] = \Pr[A \mid \text{2nd/3rd quadrant}] = \frac{3}{10}$.
  - $\Pr[A \mid C] = \Pr[A \mid \text{1st/4th quadrant}] = \frac{1}{3}$.
- Clearly, $\Pr[A \mid B] = \frac{3}{10} < \frac{1}{3} = \Pr[A \mid C]$, which violates the target inequality.

In conclusion, the inequality $\Pr[A \mid B] > \Pr[A \mid C]$ can not be derived by the given conditions.

Problem 21

Let X and Y be two random variables in a probability space $P = (\Omega, p)$. Suppose $X(\omega) \leq Y(\omega)$ for all $\omega \in \Omega$. Prove that $\mathbb{E}[X] \leq \mathbb{E}[Y]$.

Answer:

Given $X(\omega) \leq Y(\omega)$ from problem condition and $p(\omega) \geq 0$ from the definition of probability mass function, we have $(Y(\omega) - X(\omega)) \cdot p(\omega) \geq 0$ for all $\omega \in \Omega$. Therefore, by the definition of expectation, we obtain:

$$\mathbb{E}[Y - X] = \sum_{\omega \in \Omega} (Y(\omega) - X(\omega)) \cdot p(\omega) \geq 0,$$

since the sum of finitely number of non-negative terms is non-negative.

By the linearity of expectation, we have $\mathbb{E}[Y - X] = \mathbb{E}[Y] - \mathbb{E}[X] \geq 0$, which implies $\mathbb{E}[X] \leq \mathbb{E}[Y]$.

Problem 22

A rooted tree is a connected graph with no cycles together with a distinguished vertex called the root. The descendants of a vertex v are the vertices u such that v lies on the path from u to the root.

Consider the following random process. We start with a rooted tree T with n vertices, where 1 is the root. At each step, we choose uniformly at random a vertex v and delete v and all its descendants from the tree. The process stops when the tree becomes empty. Let X be the number of steps we make before the process stops. Compute $\mathbb{E}[X]$.

Answer:

We use $\text{dep}(u)$ to denote the number of vertices on the unique path from u to the root vertex 1, including both u and 1 themselves.

Let event A_i denote the event that vertex i is selected in some step of the process. This happens if and only if i is the first vertex selected among all its ancestors (including i itself). All vertices in the tree are selected uniformly at random, so for the finite set of the ancestors, each vertex is equally likely to be the first one selected. Therefore, the probability can be expressed as:

$$\Pr[A_i] = \frac{1}{\text{dep}(i)}.$$

By the definition of the process, each step selects exactly one vertex to delete, so the total number of steps X is exactly the number of vertices that are ever selected. Therefore,

$$X = \sum_{i=1}^{n} \mathbb{1}_{[A_i]}.$$

By the linearity of expectation, we have:

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^{n} \mathbb{1}_{[A_i]}\right] = \sum_{i=1}^{n} \Pr[A_i] = \sum_{i=1}^{n} \frac{1}{\text{dep}(i)}.$$

Problem 23

Suppose there are n different types of coupons. Each draw independently returns a coupon whose type is uniformly distributed over these n types.

(a) Let T_i be the number of draws until you obtain a coupon of type i for the first time. Compute $\mathbb{E}[T_i]$.

(b) Let T be the number of draws until you obtain all n types of coupons for the first time. That is, $T = \max_{i\in[n]} T_i$. Compute $\mathbb{E}[T]$ and express the result as a simple function of n up to $O(1)$ error.

Answer:

For part (a), by tail-sum formula, we can express $\mathbb{E}[T_i]$ as

$$\mathbb{E}[T_i] = \sum_{t\geq 0} \Pr[T_i > t] = \sum_{t\geq 0} \left(\frac{n-1}{n}\right)^t = \frac{1}{1-\frac{n-1}{n}} = n.$$

For part (b), we define X_i as the number of additional draws needed to obtain the i-th new coupon type, after having already collected $i-1$ distinct types. By tail-sum formula, we can express $\mathbb{E}[X_i]$ as:

$$\mathbb{E}[X_i] = \sum_{t\geq 0} \Pr[X_i > t] = \sum_{t\geq 0} \left(\frac{i-1}{n}\right)^t = \frac{1}{1-\frac{i-1}{n}} = \frac{n}{n-i+1}.$$

By the linearity of expectation, we obtain:

$$\mathbb{E}[T] = \mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} \frac{n}{n-i+1} = nH_n.$$

Using the asymptotic expansion of harmonic numbers, we can express the result $\mathbb{E}[T]$ as

$$\mathbb{E}[T] = n\left(\ln n + \gamma + \mathcal{O}\left(\frac{1}{n}\right)\right) = n\ln n + \gamma n + \mathcal{O}(1),$$

where $\gamma \approx 0.5772$, is the Euler–Mascheroni constant.